STEP2: Multi-Input Merging and Continued Propagation Arbitrary Fixed-Proportion Input/Output Recipes

1 Purpose: Merging Does Not End Propagation

STEP1 has already established that:

- A node's own production-start event is described by σ_p, ρ_p or the general $D_p^{act}(t)$;
- The supply deficit visible to a specific downstream node from a particular upstream node is described by $D_{h\to p}(t)$;
- Production delay, transportation delay, and inventories along the path are all handled by branch propagation in STEP1.

STEP2 answers only the following question:

After multiple required inputs reach the same node, how do they determine its production-start activity?

After the local activity is obtained, propagation continues: the result is passed back to STEP1 for propagation to all downstream nodes.

Thus the overall interface is

$$\boxed{\text{STEP1 propagation} \to \text{STEP2 merge} \to \text{STEP1 propagation} \to \cdots} \tag{1}$$

2 General Production Node: Arbitrary Fixed-Proportion Input/Output Recipes

Consider a production node P_p.

Its set of required inputs is

$$\mathcal{I}_p = \{1, \ldots, n_p\},$$

and its set of outputs is

$$\mathcal{O}_p = \{1, \ldots, m_p\}.$$

The fixed recipe is written as

$$\boxed{\sum_{h\in\mathcal{I}_p} a_h X_h \longrightarrow \sum_{r\in\mathcal{O}_p} b_r Y_r,} \tag{2}$$

where

$$a_h > 0, \qquad b_r > 0.$$

This is not restricted to

$$1 + 1 \to 1,$$

but allows any finite number of required inputs and any finite number of outputs, with arbitrary fixed positive input/output coefficients.

Let the initialized normal production intensity be

$$\boxed{u_p^0 \equiv u_p(0).}$$

Then the initialized normal consumption rate for input h is

$$\boxed{f_{p,h}^0 = a_h u_p^0,} \tag{3}$$

and the initialized normal completion rate for output r is

$$\boxed{g_{p,r}^0 = b_r u_p^0.} \tag{4}$$

We adopt a fixed-proportion / Leontief production recipe: All required inputs are indispensable, and neither input substitution nor dynamic recipe switching is permitted at this stage.

3 Inputs Passed from STEP1 to STEP2

For each

$$h \in \mathcal{I}_p,$$

suppose required input h corresponds to a branch whose propagation has reached the input-receiving plane of P_p.

STEP1 returns

$$\boxed{D_{h\to p}(t).} \tag{5}$$

Its meaning is:

$$\boxed{\begin{array}{c} D_{h\to p}(t) = \\ \text{cumulative equivalent normal-production time missing} \\ \text{for input } h, \text{ as seen by } P_p \text{ by time } t \end{array}}$$

Therefore the corresponding cumulative material deficit is

$$\boxed{a_h u_p^0 D_{h\to p}(t).} \tag{6}$$

All

$$D_{h\to p}(t)$$

must already refer to the same observation plane:

$$\boxed{\begin{array}{c} \text{the production-start plane where } P_p \text{ can access} \\ \text{all required inputs and decide whether to initiate production} \end{array}}$$

STEP2 does not recalculate anything already covered by STEP1, including production delay, transportation delay, or inventory coverage.

4 Single-Window Input Branches and General Input Branches

If input branch h has only one effective supply-interruption window

$$[t^{\text{stop}}_{h\to p}, t^{\text{rec}}_{h\to p}),$$

then

$$\boxed{D_{h\to p}(t) = \min\left\{(t - t^{\text{stop}}_{h\to p})_+, \Delta_{h\to p}\right\},} \tag{7}$$

where

$$\boxed{\Delta_{h\to p} = \left(t^{\text{rec}}_{h\to p} - t^{\text{stop}}_{h\to p}\right)_+.} \tag{8}$$

However, in a general DAG an input branch may already contain multiple disjoint deficits, so the main interface between STEP1 and STEP2 is always the complete function

$$\boxed{D_{h\to p}(t),}$$

rather than merely a pair

$$(t^{\text{stop}}, t^{\text{rec}}).$$

5 Why Merging Is Not a Simple Union of Windows

If any required input temporarily prevents P_p from producing, other required inputs will also stop being consumed by the production process.

These nonbinding inputs may keep arriving and be stored during the shutdown, creating additional buffers for later periods.

Thus, shortage windows computed independently for different input branches must not be simply unioned, nor should their shortage durations be directly added.

The correct objects are the individual inputs'

$$D_{h\to p}(t),$$

which are compared as cumulative constraints at the production-start plane.

6 Main STEP2 Theorem: Upper Envelope of Required-Input Constraints

By time t, if the production node had operated continuously at its normal intensity, the cumulative consumption of input h would have been

$$a_h u_p^0 t.$$

Because this input branch has a cumulative shortfall of

$$a_h u_p^0 D_{h\to p}(t),$$

the maximum cumulative production activity that this input can support by time t is

$$\frac{a_h u_p^0 t - a_h u_p^0 D_{h\to p}(t)}{a_h}.$$

Hence

$$\boxed{\frac{a_h u_p^0 t - a_h u_p^0 D_{h\to p}(t)}{a_h} = u_p^0\left[t - D_{h\to p}(t)\right].} \tag{9}$$

Let the cumulative production activity initiated by node P_p up to time t be

$$\boxed{U_p(t) = \int_0^t u_p(s)\, ds.} \tag{10}$$

Since all required inputs must be simultaneously satisfied,

$$U_p(t) \le u_p^0\,[t - D_{h\to p}(t)] \qquad \forall h \in \mathcal{I}_p.$$

Under the current greedy baseline, the node produces at normal intensity whenever materials permit, so actual cumulative activity takes the tightest constraint:

$$\begin{aligned} U_p(t) &= \min_{h\in\mathcal{I}_p} u_p^0\,[t - D_{h\to p}(t)] \\ &= u_p^0\left[t - \max_{h\in\mathcal{I}_p} D_{h\to p}(t)\right]. \end{aligned} \tag{11}$$

Define the node's own cumulative production-start deficit

$$\boxed{D_p^{act}(t) = \max_{h\in\mathcal{I}_p} D_{h\to p}(t).} \tag{12}$$

Then

$$\boxed{U_p(t) = u_p^0\left[t - D_p^{act}(t)\right].} \tag{13}$$

Away from the switching points of events,

$$\boxed{u_p(t) = u_p^0\left[1 - \dot{D}_p^{act}(t)\right].} \tag{14}$$

Under the complete-production / complete-shutdown baseline,

$$\dot{D}_p^{act}(t) \in \{0, 1\}$$

holds almost everywhere.

7 Why Arbitrary Input Coefficients a_h Do Not Change the max Structure

Equation (9) shows that each required input's material deficit is divided by its own recipe coefficient a_h, so that all constraints are expressed in the same units:

$$\boxed{\text{Equivalent time of normal production activity.}}$$

Thus although

$$a_1, a_2, \ldots, a_{n_p}$$

may be arbitrarily different, after normalization each required input yields an

$$t - D_{h\to p}(t)$$

activity constraint of this form.

Therefore

$$\boxed{\text{arbitrary fixed-proportion required-input coefficients} \quad\Longrightarrow\quad D_p^{act}(t) = \max_h D_{h\to p}(t).} \tag{15}$$

The input coefficients a_h still affect normal flow rates,

$$f_{p,h}^0 = a_h u_p^0$$

and thereby material quantities and inventory coverage, but not the normalized max-merge form.

8 Arbitrary Output Coefficients and Branch Allocation

After merging, P_p has just one common production-start activity

$$u_p(t), \qquad D_p^{act}(t).$$

The completion rate of output r is

$$g_{p,r}(t) = b_r u_p(t).$$

If that output is then allocated at a fixed proportion

$$\rho_{p,r\to q}$$

to downstream node P_q, the normal flow rate of that output branch is

$$\boxed{f^0_{p,r\to q} = \rho_{p,r\to q} b_r u^0_p.} \tag{16}$$

Thus different outputs may have arbitrary fixed positive coefficients and different fixed allocation ratios. These coefficients scale material quantities and enter each branch's inventory coverage, but all outputs share the same local production activity

$$D_p^{act}(t).$$

9 Reading P_p's Own Stop/Recovery Events from the Merged Result

From

$$D_p^{act}(t)$$

read all maximal connected intervals satisfying

$$\dot{D}_p^{act}(t) = 1$$

as:

$$\boxed{I_p^{act,\ell} = [\sigma_p^\ell, \rho_p^\ell), \qquad \ell = 1, \ldots, L_p.} \tag{17}$$

These intervals represent

$$\boxed{\text{production-start gaps when } P_p \text{ itself cannot initiate new production.}}$$

These

$$\sigma_p^\ell, \rho_p^\ell$$

still are not stop/recovery events propagated to an arbitrary downstream node; the specific downstream node must be processed again through STEP1.

If the merged result has only one internal window

$$[\sigma_p, \rho_p),$$

it reduces exactly to the single-window input format of STEP1.

10 How to Continue Propagating Shocks Downstream After Merging

Suppose an output of P_p supplies downstream node P_q along a direct branch.

STEP2's output is not an endpoint; it is

$$\boxed{D_p^{act}(t),}$$

and immediately becomes the input to the next STEP1 operation.

Define the output branch's

$$\Theta_{p\to q} = \tau_p^{prod} + \sum_{e\in\mathcal{E}_{p\to q}} \tau_e^{trans},$$

and

$$C_{p\to q} = \sum_{m\in\mathcal{M}_{p\to q}} \frac{q_m(0)}{f_m^{out,0}}.$$

Then

$$\boxed{D_{p\to q}(t) = \mathcal{T}_{p\to q}[D_p^{act}](t) = \left[D_p^{act}(t-\Theta_{p\to q}) - C_{p\to q}\right]_+.} \tag{18}$$

Therefore

$$\boxed{\{D_{h\to p}\}_h \overset{\text{STEP2}}{\longrightarrow} D_p^{act} \overset{\text{STEP1}}{\longrightarrow} D_{p\to q}.} \tag{19}$$

11 Recovering the First Version's Symmetric Downstream Formulas for a Single Merged Window

If P_p has only one production-start gap after merging in STEP2,

$$[\sigma_p, \rho_p),$$

and the gap is long enough to penetrate all inventory on the $P_p \to P_q$ branch, i.e.,

$$\rho_p - \sigma_p > C_{p\to q},$$

STEP1 immediately gives

$$\boxed{\begin{aligned} t_{p\to q}^{\text{stop}} &= \sigma_p + \Theta_{p\to q} + C_{p\to q}, \\ t_{p\to q}^{\text{rec}} &= \rho_p + \Theta_{p\to q}. \end{aligned}} \tag{20}$$

Thus even when P_p itself merges multiple inputs, as long as the local result is a single window, the same formulas still hold exactly for any downstream node:

$$\boxed{\text{stop} = \text{local stop} + \text{delay} + \text{inventory}, \qquad \text{recovery} = \text{local recovery} + \text{delay}.} \tag{21}$$

If

$$\rho_p - \sigma_p \le C_{p\to q},$$

then the output branch's inventory fully absorbs the internal shutdown, and no effective supply interruption occurs at P_q.

12 Continued Propagation of Multi-Window Merge Results

If

$$D_p^{act}(t)$$

contains multiple disjoint increasing intervals, one cannot apply the initial inventory $C_{p\to q}$ separately to each

$$[\sigma_p^{\ell}, \rho_p^{\ell})$$

interval.

The correct procedure is to evaluate the entire cumulative function once:

$$\boxed{D_{p\to q}(t) = [D_p^{act}(t - \Theta_{p\to q}) - C_{p\to q}]_+.} \tag{22}$$

Then, at the receiving plane of P_q, read

$$\boxed{[t_{p\to q}^{\mathrm{stop},j}, t_{p\to q}^{\mathrm{rec},j}) = \text{the } j\text{-th maximal connected interval with } \dot{D}_{p\to q}(t) = 1.} \tag{23}$$

This prevents the same downstream inventory from being erroneously reinitialized for multiple windows.

13 One Merge Node Can Cause Different Shocks at Different Downstream Nodes

If P_p supplies

$$P_{q_1}, P_{q_2}, \ldots, P_{q_s},$$

then these nodes share the same

$$D_p^{act}(t),$$

but each output branch has its own

$$\Theta_{p\to q_j}, \qquad C_{p\to q_j}.$$

Therefore

$$\boxed{D_{p\to q_j}(t) = \left[D_p^{act}(t - \Theta_{p\to q_j}) - C_{p\to q_j}\right]_+, \qquad j = 1, \ldots, s.} \tag{24}$$

Consequently, the following can occur:

$$P_p \text{ itself has already shut down,}$$

yet one downstream node with ample inventory has felt no supply interruption, while another with little inventory has already experienced a shortage.

Therefore

$$\boxed{\text{The local activity event is shared by the node;the downstream-visible event is branch-specific.}} \tag{25}$$

14 Two-Input Example: Shutdown, Recovery, and Another Shutdown After Merging

Consider

$$A + B \longrightarrow Y.$$

Suppose STEP1 has already propagated both upstream branches to Y's input-receiving plane:

$$D_{A\to Y}(t) = \begin{cases} 0, & t < 2, \\ t-2, & 2 \le t < 5, \\ 3, & t \ge 5, \end{cases}$$

and

$$D_{B\to Y}(t) = \begin{cases} 0, & t < 4, \\ t-4, & 4 \le t < 8, \\ 4, & t \ge 8. \end{cases}$$

STEP2 merge:

$$\boxed{D_Y^{act}(t) = \max\left\{D_{A\to Y}(t), D_{B\to Y}(t)\right\}.}$$

We obtain

$$D_Y^{act}(t) = \begin{cases} 0, & t < 2, \\ t-2, & 2 \le t < 5, \\ 3, & 5 \le t < 7, \\ t-4, & 7 \le t < 8, \\ 4, & t \ge 8. \end{cases}$$

Thus Y's own production-start gaps are

$$\boxed{[2,5) \quad \text{and} \quad [7,8).}$$

To determine how these internal gaps affect some downstream node Z, we must not stop at STEP2, but must continue with STEP1:

$$\boxed{D_{Y\to Z}(t) = \left[D_Y^{act}(t - \Theta_{Y\to Z}) - C_{Y\to Z}\right]_+.} \tag{26}$$

Only the increasing intervals read from $D_{Y\to Z}$ represent the true events observed by Z:

$$t_{Y\to Z}^{\text{stop},j}, \qquad t_{Y\to Z}^{\text{rec},j}.$$

15 Correspondence with "Production Status of Plant p on Day k"

The STEP2 result

$$D_p^{act}(t)$$

directly determines the production status of P_p:

$$\boxed{\frac{u_p(t)}{u_p^0} = 1 - \dot{D}_p^{act}(t).} \tag{27}$$

If time is discretized by day with no additional intraday switching on day k, define

$$\boxed{x_p[k] = 1 - \left(D_p^{act}(k+1) - D_p^{act}(k)\right).} \tag{28}$$

Under the binary baseline,

$$x_p[k] \in \{0, 1\},$$

they respectively indicate normal production or complete shutdown on that day.

Thus, STEP2's max merge does not merely give the "merge result"; it also directly gives the original question's target after the merge node:

$$\boxed{\text{Production status at every time / on every day.}}$$

16 Unified Recursion for an Arbitrary DAG

For any production node P_p, the calculation follows this fixed order.

Input Side: Apply STEP1 First to Every Required-Input Branch

For each

$$h \in \mathcal{I}_p,$$

calculate

$$\boxed{D_{h\to p}(t) = \mathcal{T}_{h\to p}[D_h^{act}](t).} \tag{29}$$

At the Node: STEP2 Performs the Required-Input Merge

$$\boxed{D_p^{act}(t) = \max_{h\in\mathcal{I}_p} D_{h\to p}(t).} \tag{30}$$

And from

$$\boxed{u_p(t) = u_p^0[1 - \dot{D}_p^{act}(t)]} \tag{31}$$

obtain the local production status.

Output Side: Apply STEP1 Again to Every Downstream Branch

For each direct downstream P_q,

$$\boxed{D_{p\to q}(t) = \mathcal{T}_{p\to q}[D_p^{act}](t).} \tag{32}$$

Thus the entire DAG repeatedly uses just two operations:

$$\boxed{\text{Along a branch: delay shift + inventory absorption,}}$$

and

$$\boxed{\text{At a production node: required-input max merge.}}$$

This can be written as

$$\boxed{\mathcal{T} \to \max \to \mathcal{T} \to \max \to \cdots .} \tag{33}$$

17 Consistency with Single-Input Nodes

If

$$|\mathcal{I}_p| = 1,$$

then

$$\boxed{D_p^{act}(t) = D_{h\to p}(t).} \tag{34}$$

Thus STEP2 automatically reduces to the identity mapping. The entire network then requires only STEP1 operator composition along linear chains, reducing to the original segment-by-segment cumulative formulas.

18 Inventory Ownership and Avoiding Double Counting

We must enforce

> One physical inventory belongs to exactly one propagation segment.

If an inventory has already been included in an input branch's

$$C_{h\to p},$$

STEP2 must not subtract it again as a new buffer.

If P_p has separate input-side local storage, it should be represented explicitly as an inventory node M in the topology, with clear ownership by a particular STEP1 propagation segment.

Similarly, if finished-goods storage absorbs an output deficit of P_p, it should be handled as an inventory node in output-side STEP1, not hidden within production delay or the STEP2 merge.

19 Applicability Limits of the Current Closed-Form max Rule

The equation

$$D_p^{act}(t) = \max_h D_{h\to p}(t)$$

relies on the following baseline:

- All inputs are fixed-proportion required inputs;
- Inputs cannot directly substitute for one another;
- Production recipe coefficients a_h, b_r are fixed;
- Normally, the node operates at u_p^0 whenever materials permit;
- If one input causes a joint shutdown, other arriving inputs can be stored;
- Storage-capacity limits, losses, scrapping, and shelf-life constraints are not yet considered;
- No active restocking, overproduction, or rerouting;
- The current focus is complete shutdown / complete recovery.

Thus "arbitrary input/output ratios" here means precisely

> a multi-input / multi-output Leontief recipe with fixed positive coefficients,

not an arbitrary substitutable, switchable, or nonlinear production technology.

If nonbinding inputs cannot continue to be stored during a joint shutdown, for example, if local storage capacity is binding, then the simple

$$D_p^{act}(t) = \max_h D_{h\to p}(t)$$

may fail, and explicit inventory-conservation equations are needed.

If partial shortages are allowed, the rule can be generalized to

$$0 \le \dot{D}(t) \le 1,$$

In that case

$$u_p(t) = u_p^0[1 - \dot{D}_p^{act}(t)]$$

naturally represents partial production.

20 Final STEP2 Interface

STEP2 can be summarized as

$$\boxed{\underbrace{\{D_{h\to p}(t)\}_{h\in\mathcal{I}_p}}_{\text{required-input shortages propagated to } P_p \text{ by STEP1}} \xrightarrow{\max} \underbrace{D_p^{act}(t)}_{\text{local production-start activity gap of } P_p}} \tag{35}$$

where

$$\boxed{D_p^{act}(t) = \max_{h\in\mathcal{I}_p} D_{h\to p}(t).}$$

Then for every direct downstream P_q:

$$\boxed{D_p^{act}(t) \xrightarrow{\text{STEP1}} D_{p\to q}(t) = [D_p^{act}(t-\Theta_{p\to q}) - C_{p\to q}]_+.} \tag{36}$$

The final closed loop is therefore

$$\boxed{\begin{array}{l}\text{upstream local activity} \to \text{STEP1 propagation} \to \text{required-input shortages} \\ \to \text{STEP2 max merge} \to \text{local production activity} \to \text{STEP1 propagation}\end{array}}$$

For a single window, one can always return to the first version's symmetric formulas; For multiple windows, preserve the entire $D(t)$ function for continued recursion.